ĐẠI HỌC TÔN ĐỨC THẮNG
TON DUC THANG UNIVERSITY

Introduction to Machine Learning

LÊ ANH CƯỜNG
Ton Duc Thang University
2024

Content

1. What is AI and what is ML?
2. How ML helps or why learn ML?
3. The general diagram of machine learning model building; what is model in ML?
4. Key components of ML
5. Type of machine learning methods
6. History of ML

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AI and ML

- What is Artificial Intelligence (AI)?
- What is Machine Learning (ML)?
- Relationship between AI and ML?

AI

- Artificial Intelligence (AI) is a field of study focused on methods and techniques that enable computers and machines to simulate human intelligence, demonstrated through capabilities such as:
  - human learning,
  - comprehension,
  - problem solving,
  - decision making,
  - creativity and autonomy.

- Trí tuệ nhân tạo (AI) là lĩnh vực nghiên cứu phương pháp, kỹ thuật cho phép máy tính và máy móc mô phỏng khả năng học tập, hiểu biết, giải quyết vấn đề, ra quyết định, sáng tạo và tự chủ của con người.

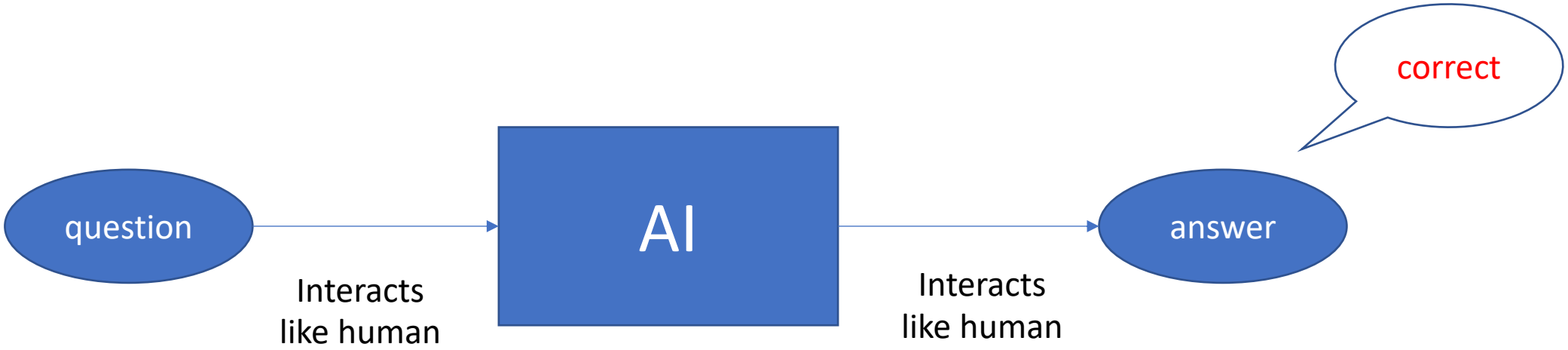
Machine Learning

- Machine learning involves creating models by training an algorithm to make predictions or decisions based on data.
- It encompasses a broad range of techniques that enable computers to learn from and make inferences based on data without being explicitly programmed for specific tasks.

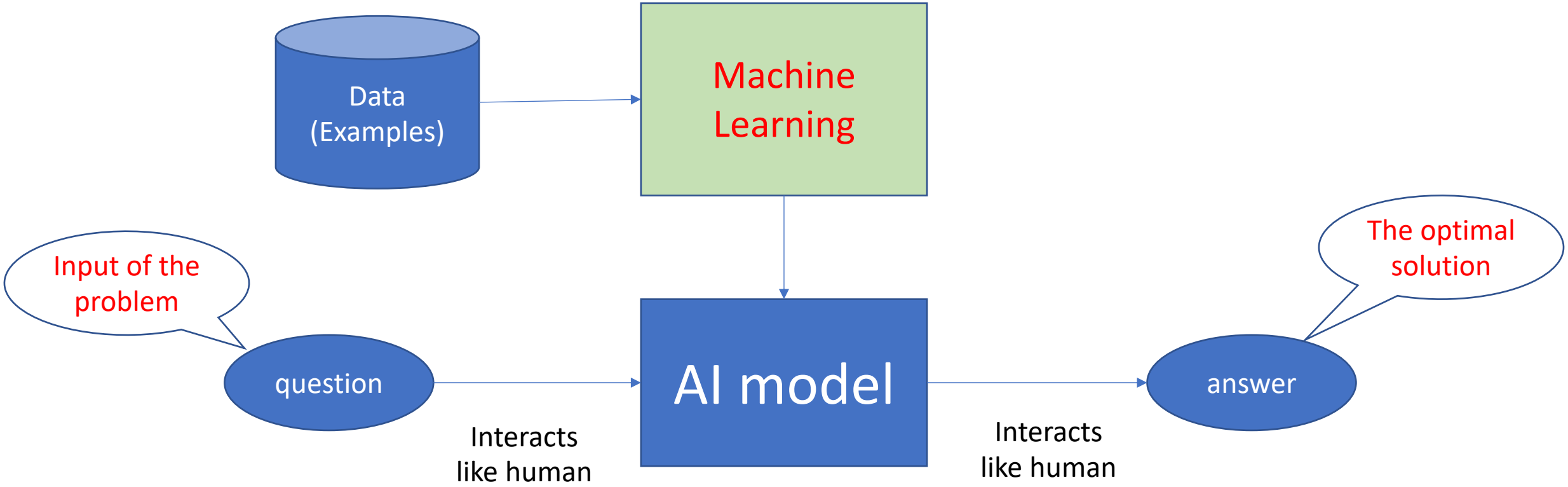
- Học máy liên quan đến việc tạo ra các mô hình bằng cách đào tạo một thuật toán để đưa ra dự đoán hoặc quyết định dựa trên dữ liệu.
- Nó bao gồm một loạt các kỹ thuật cho phép máy tính học hỏi và đưa ra suy luận dựa trên dữ liệu mà không cần được lập trình rõ ràng cho các tác vụ cụ thể.

Narrow AI

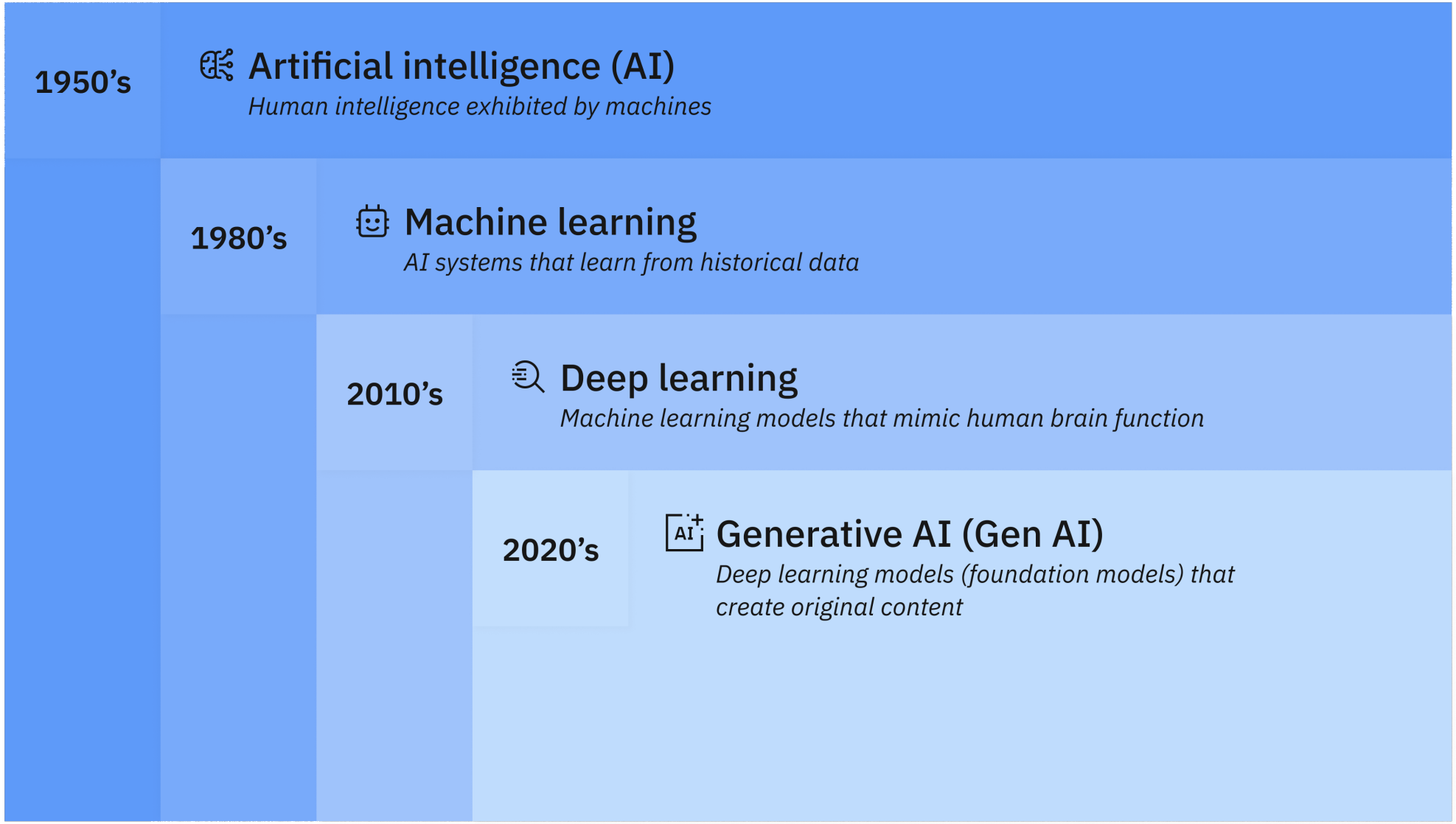
- AI is expected as experts:
  - AI offers *optimal* solutions to difficult/complicated problems
  - AI solves predictive and focasting problems (uncertainty)
  - Interacts like human



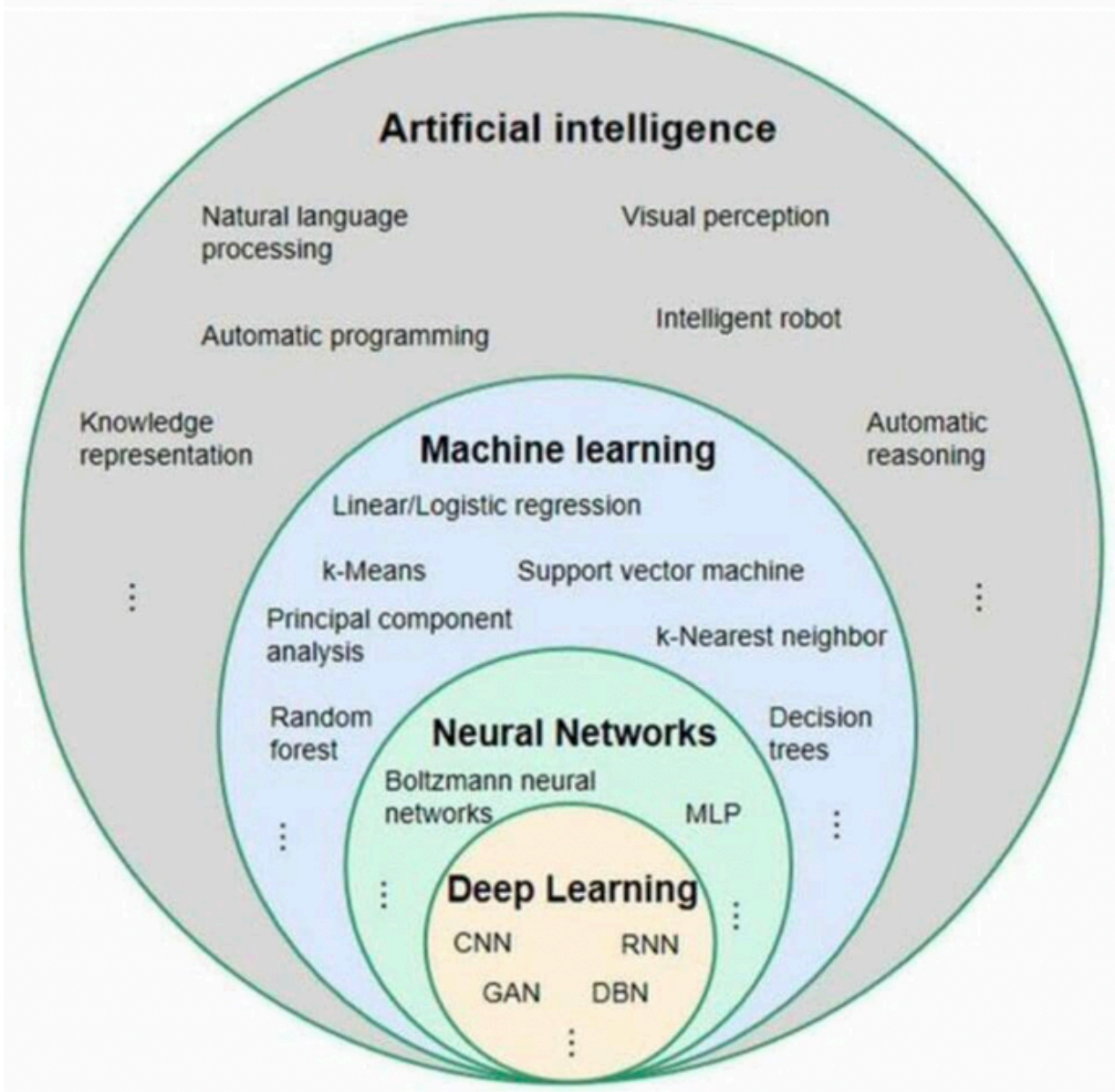
What is Machine Learning?



AI vs ML?



https://www.ibm.com/topics/artificial-intelligence

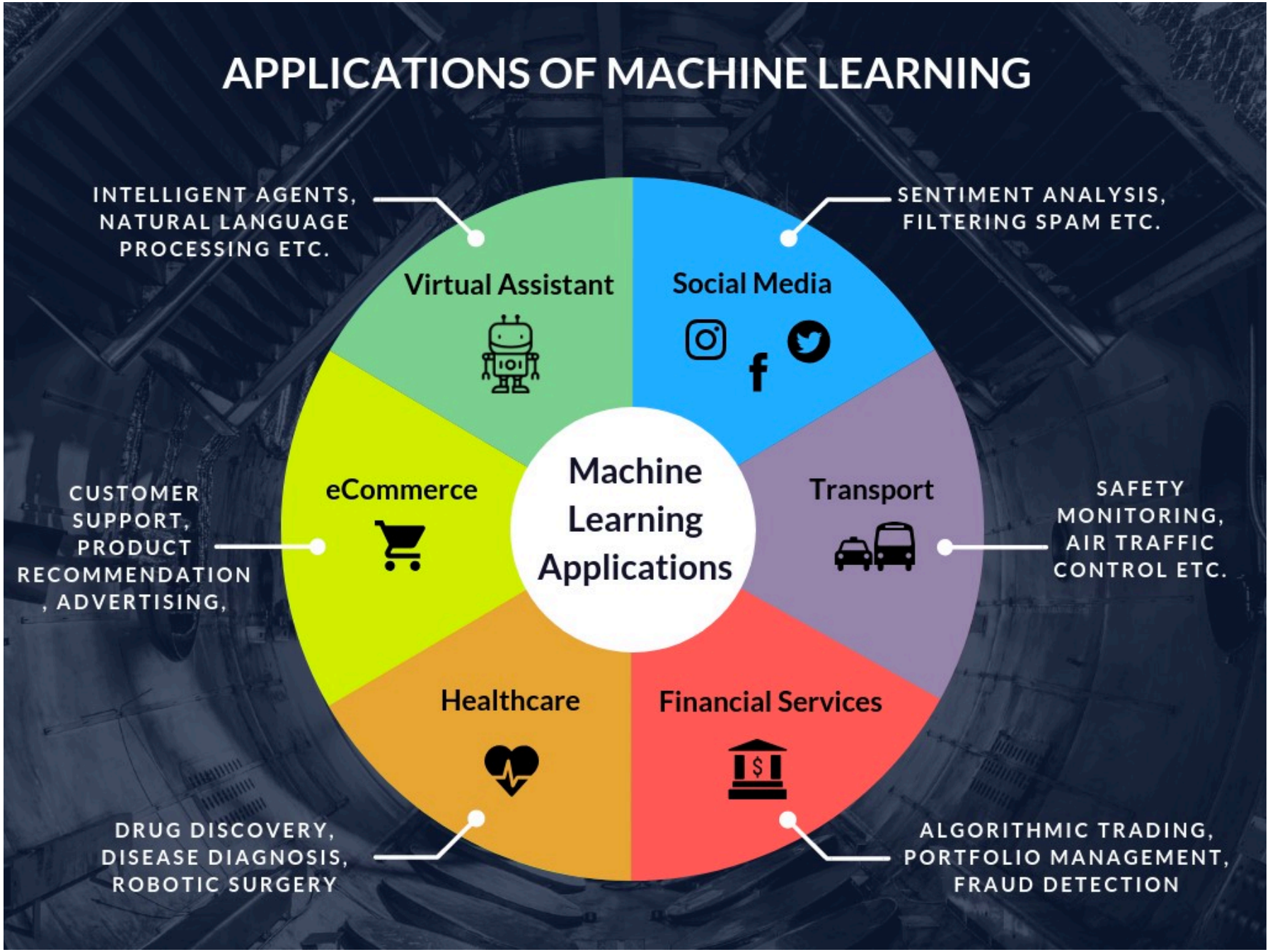

Artificial intelligence
Natural language processing
Visual perception
Automatic programming
Intelligent robot
Knowledge representation
Machine learning
Automatic reasoning
Linear/Logistic regression
k-Means
Support vector machine
Principal component analysis
k-Nearest neighbor
Random forest
Neural Networks
Decision trees
Boltzmann neural networks
MLP
Deep Learning
CNN
RNN
GAN
DBN

ML vs explicit Algorithms?

- ML: It encompasses a broad range of techniques that enable computers to learn from and make inferences based on data without being explicitly programmed for specific tasks
- ML is a *black box* model, it can not explain clearly (why, meaning) the result is generated.

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APPLICATIONS OF MACHINE LEARNING
INTELLIGENT AGENTS, NATURAL LANGUAGE PROCESSING ETC.
Virtual Assistant
SENTIMENT ANALYSIS, FILTERING SPAM ETC.
Social Media
CUSTOMER SUPPORT, PRODUCT RECOMMENDATION , ADVERTISING,
eCommerce
Machine Learning Applications
Transport
SAFETY MONITORING, AIR TRAFFIC CONTROL ETC.
Healthcare
Financial Services
DRUG DISCOVERY, DISEASE DIAGNOSIS, ROBOTIC SURGERY
ALGORITHMIC TRADING, PORTFOLIO MANAGEMENT, FRAUD DETECTION

Domains of applying AI

- Retail
  - AI provides virtual shopping capabilities that offer personalized recommendations and discuss purchase options with the consumer.
- Banking
  - Artificial Intelligence enhances the speed, precision and effectiveness of human efforts. In financial institutions, AI techniques can be used to identify which transactions are likely to be fraudulent, adopt fast and accurate credit scoring, as well as automate manually intense data management tasks.
- Games
  - AI in games is about planning. The artificial agent must decide on the next best action(s) to perform depending on the state of the game.

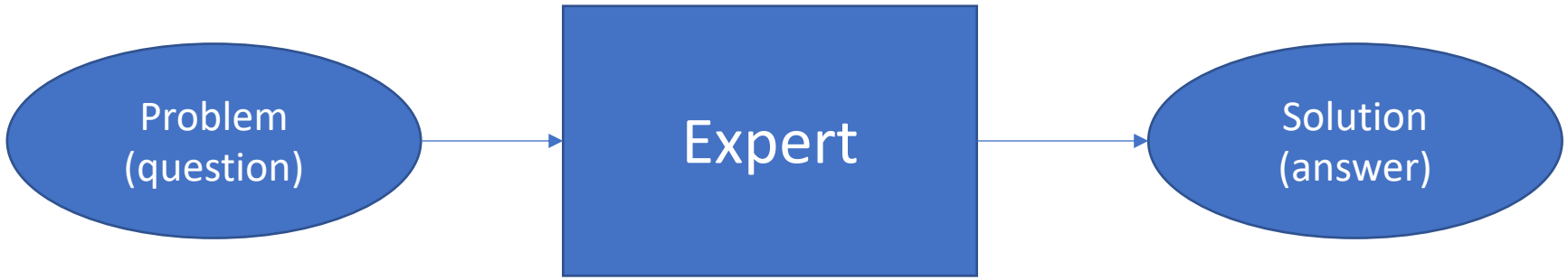
Domains of applying AI

- Transportation
  - autonomous cars; navigation system; monitoring traffic lights;
- Logistics and Warehouse Management
  - Logistics and warehouses management are two very complex problems. They depends on many parameters difficult to model, some of them even time-dependent.
- Machine Translation
- Virtual Assistant
- Computer Vision
- Healthcare
- …

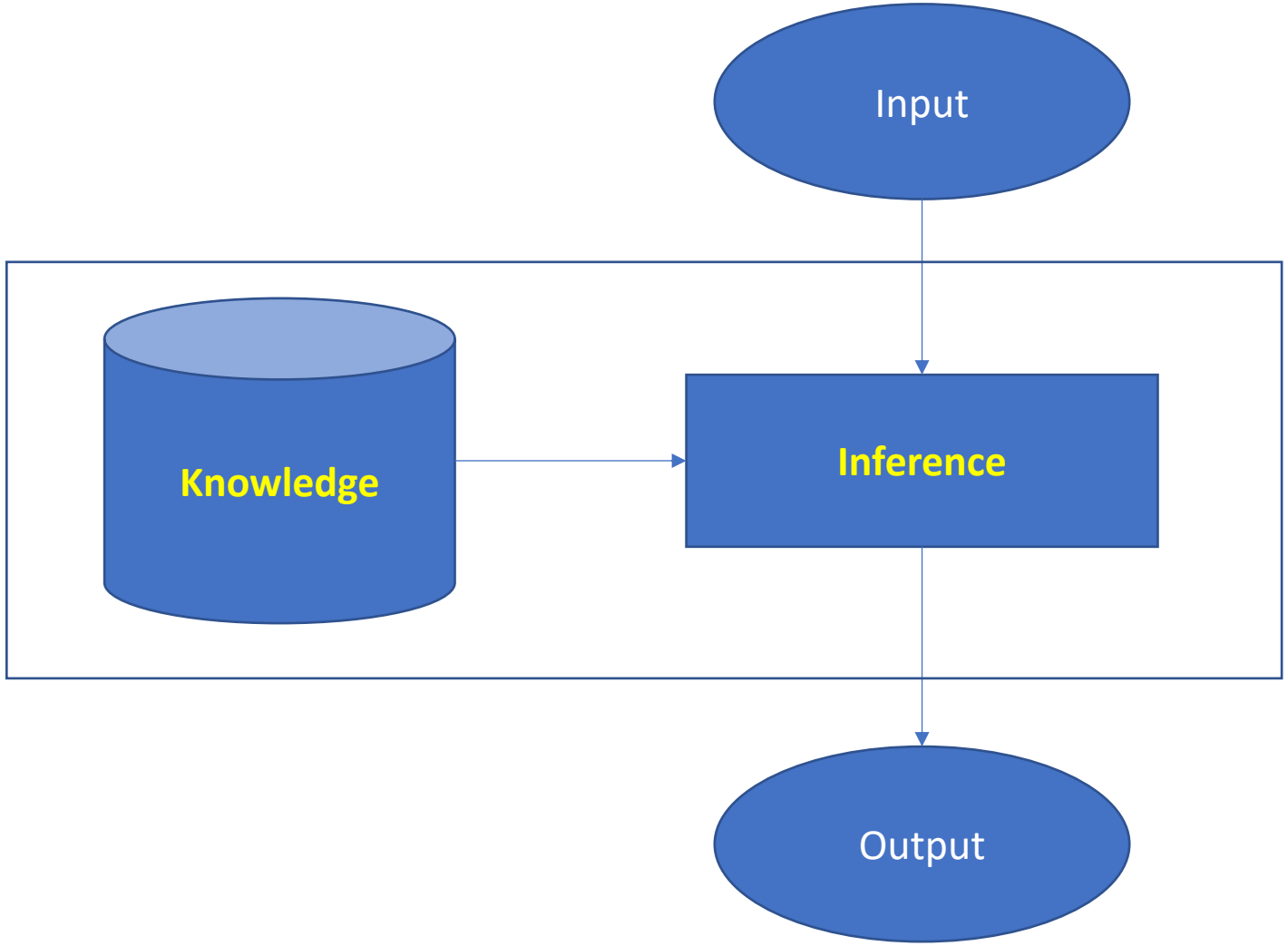
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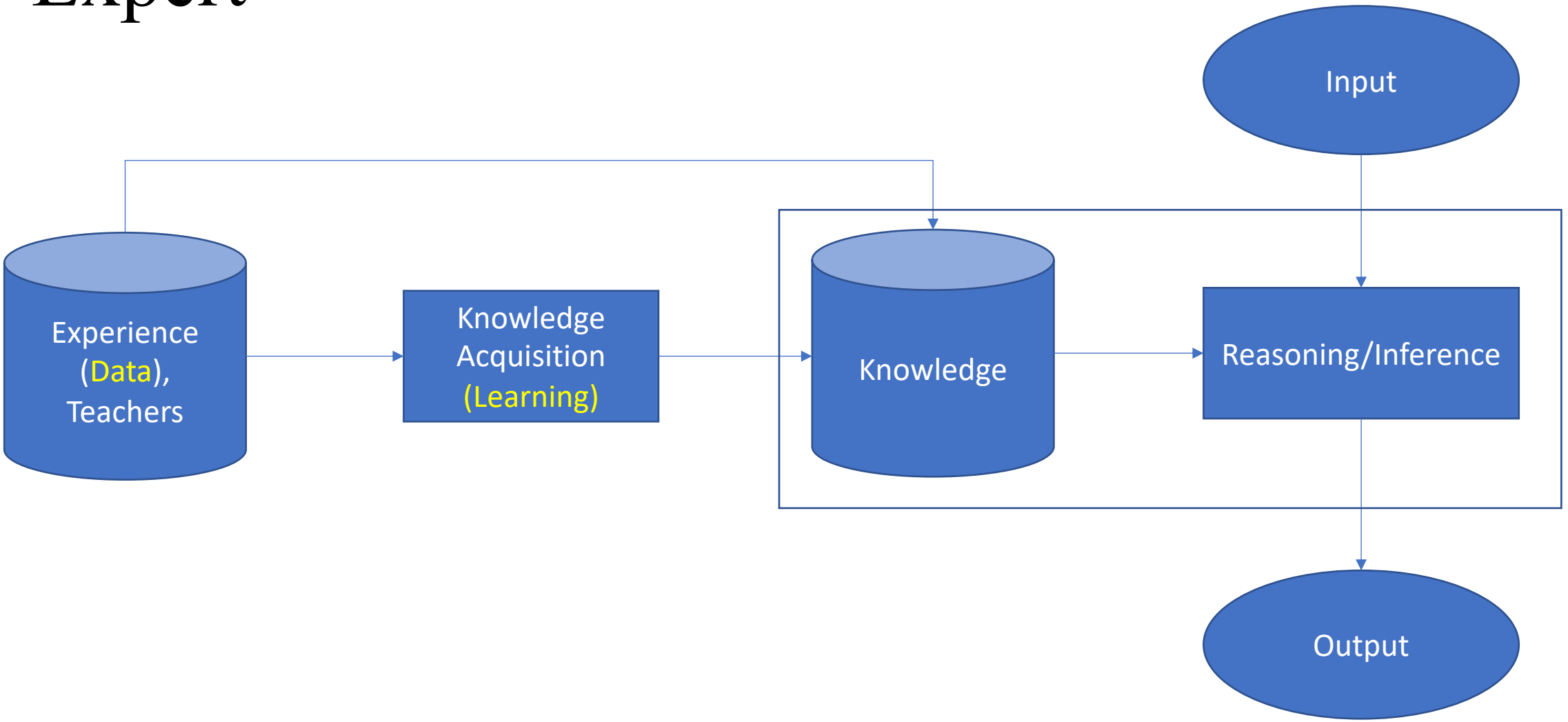
How does an Expert solve problems?



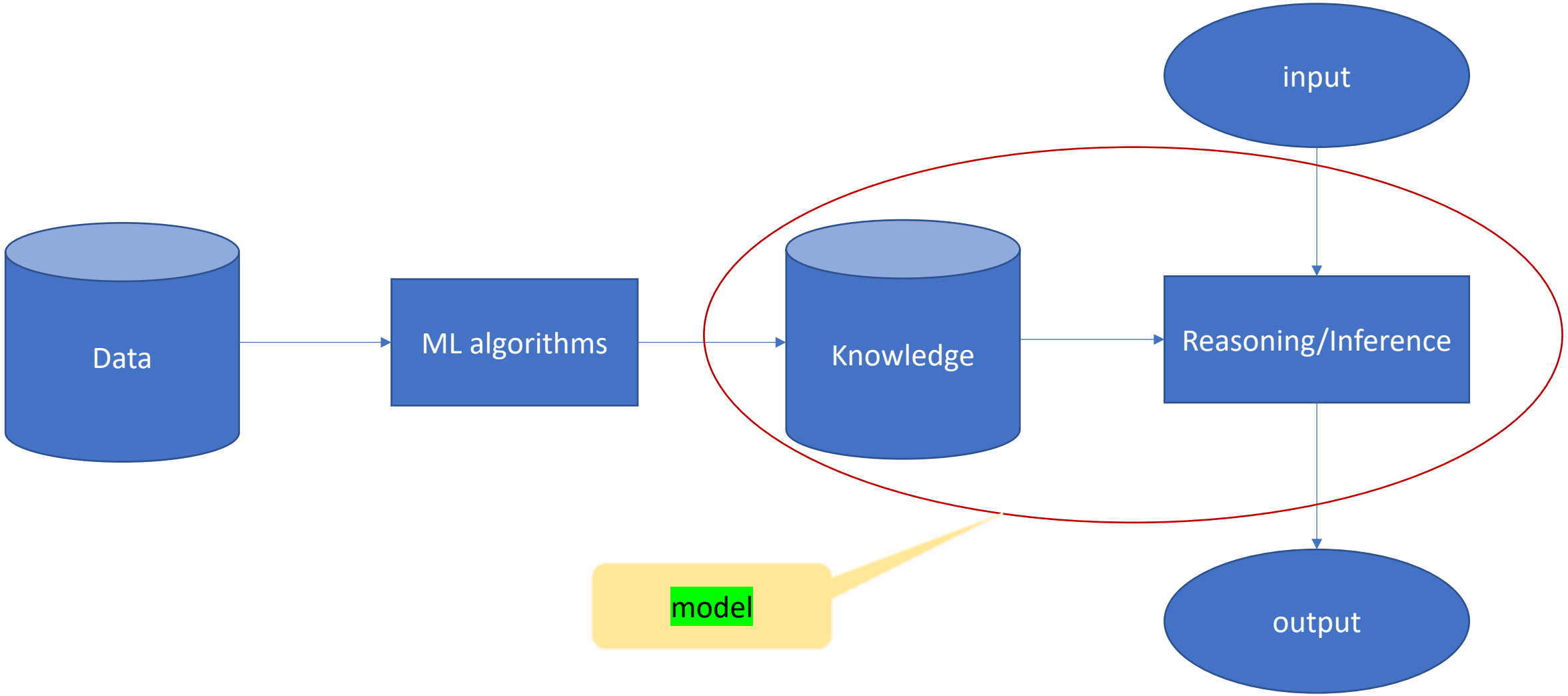
Expert



Expert


Input
Experience
(Data),
Teachers
Knowledge
Acquisition
(Learning)
Knowledge
Reasoning/Inference
Output

A General Paradigm of ML Systems

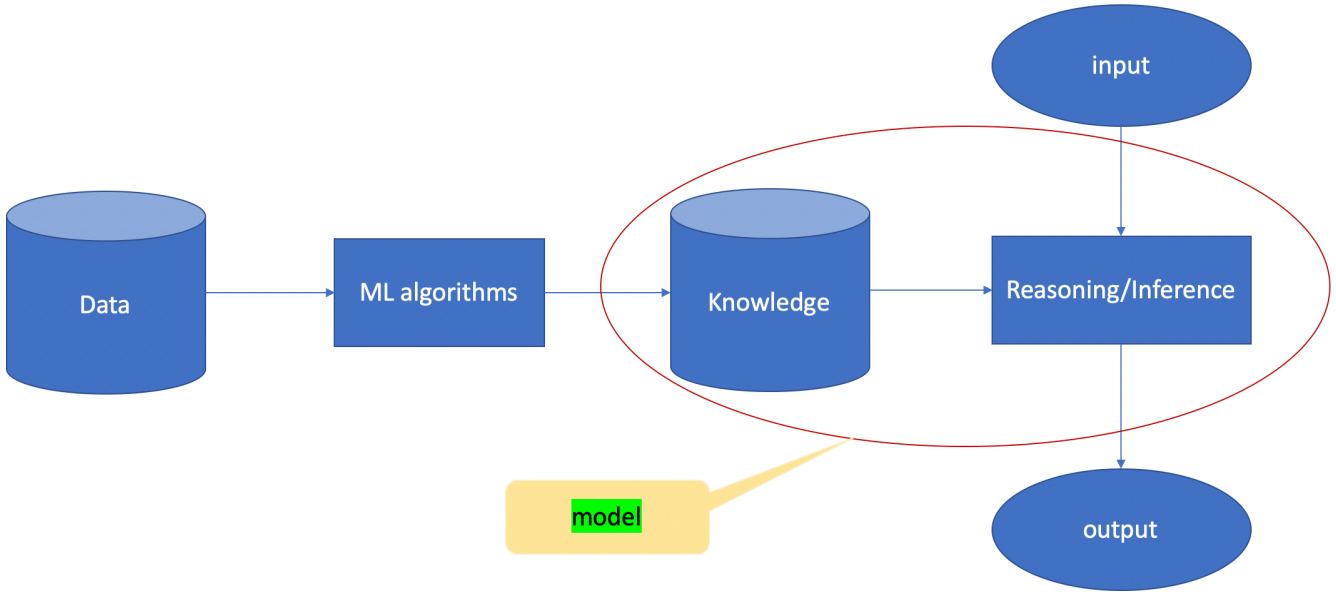


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7. Machine learning and other fields

What are ML components in AI systems?

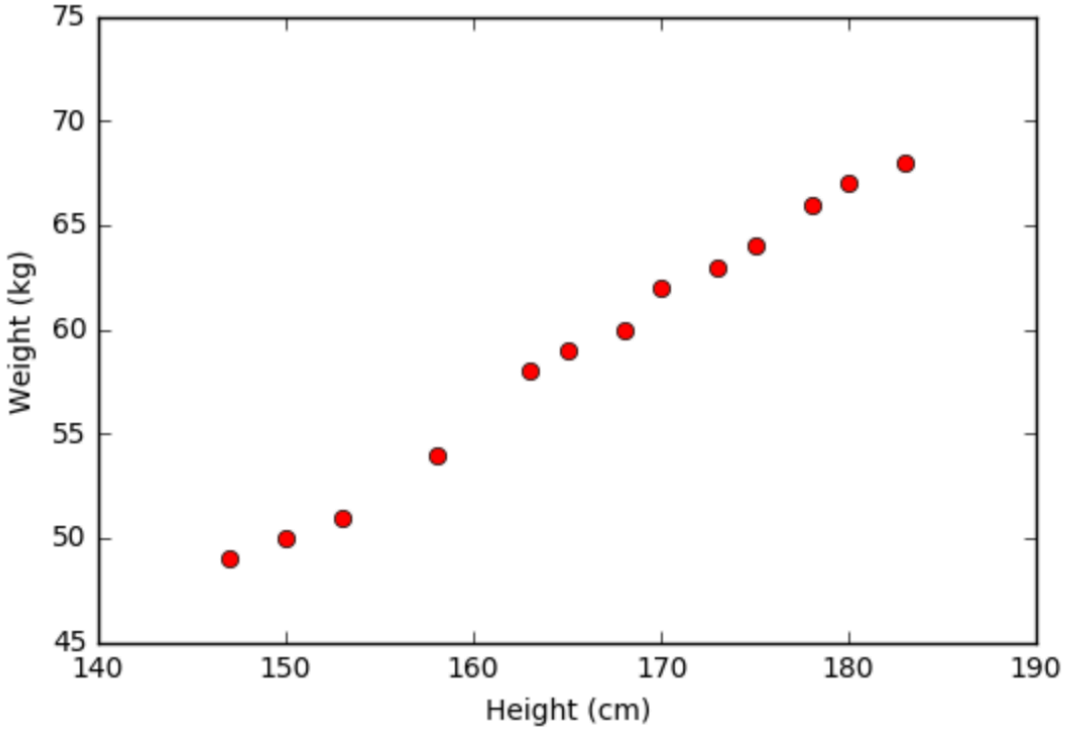
- Data
- Model
- Learning algorithms



An example of a machine learning problem

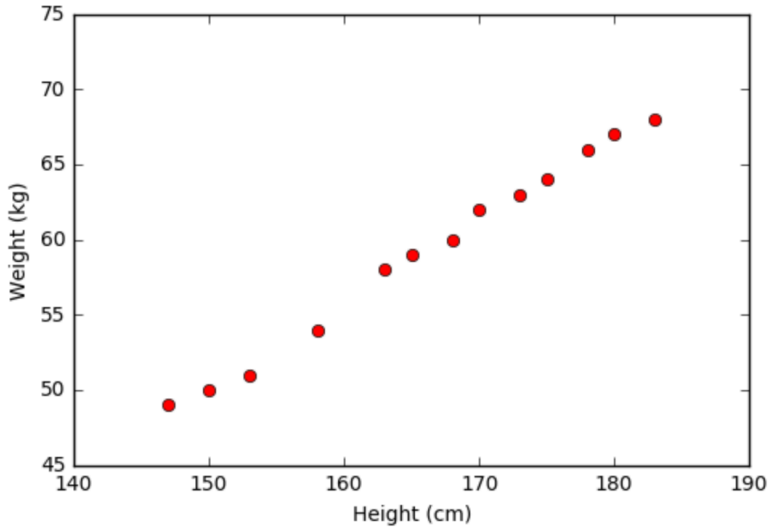
Height (cm)	Weight (kg)	Height (cm)	Weight (kg)
147	49	168	60
150	50	170	72
153	51	173	63
155	52	175	64
158	54	178	66
160	56	180	67
163	58	183	68
165	59		

An example of a machine learning problem

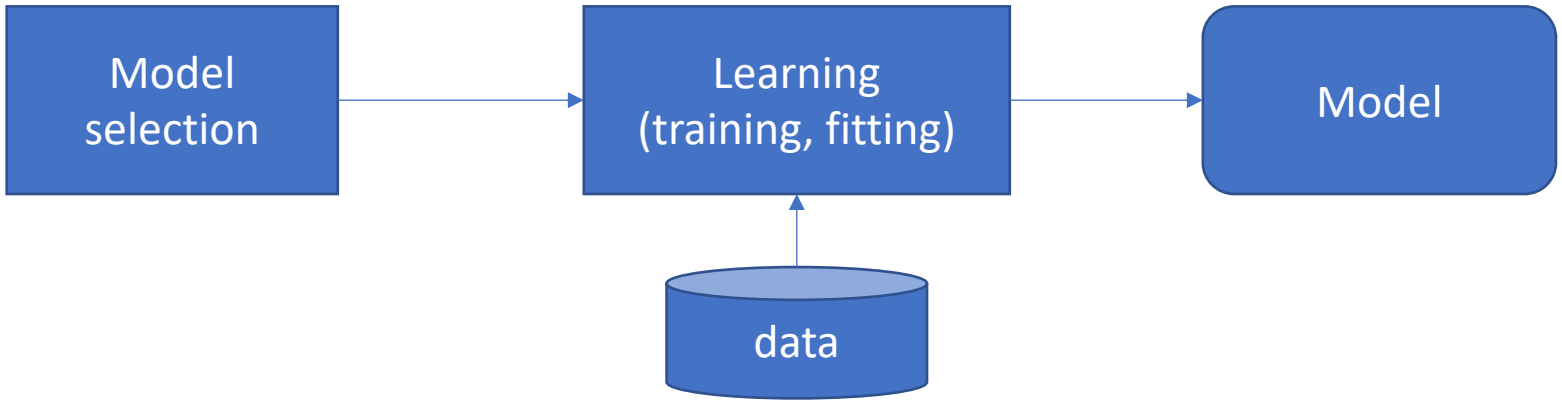


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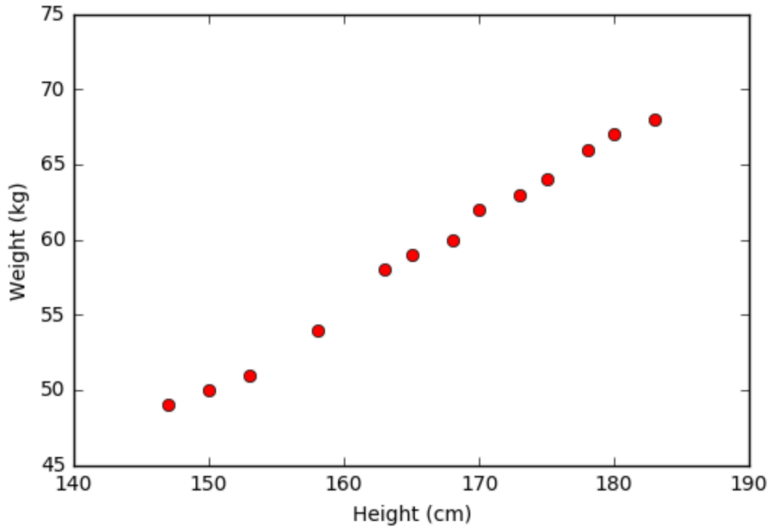
An example of a machine learning problem



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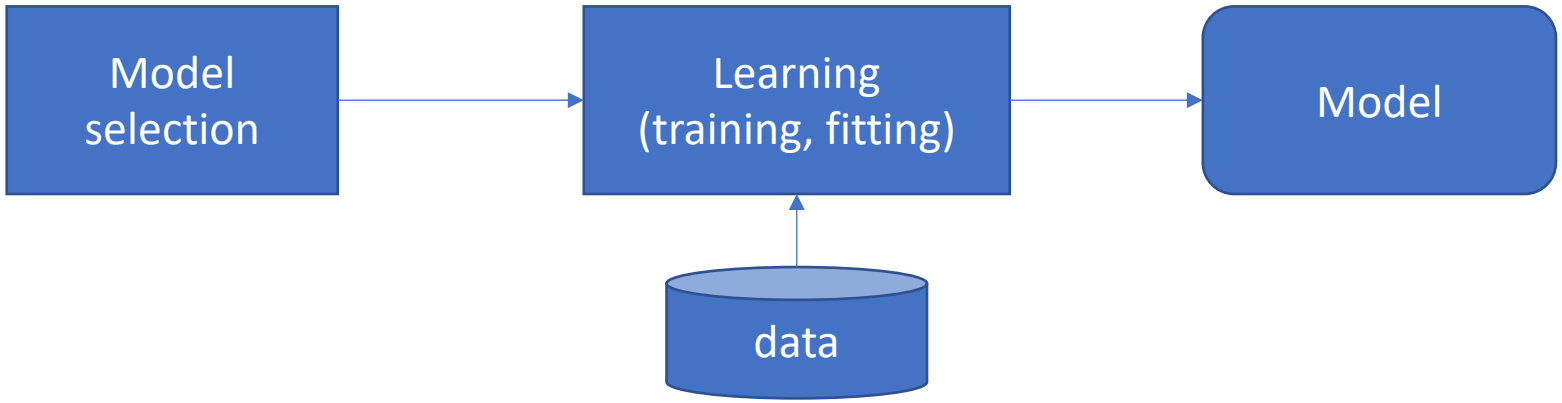


An example of a machine learning problem



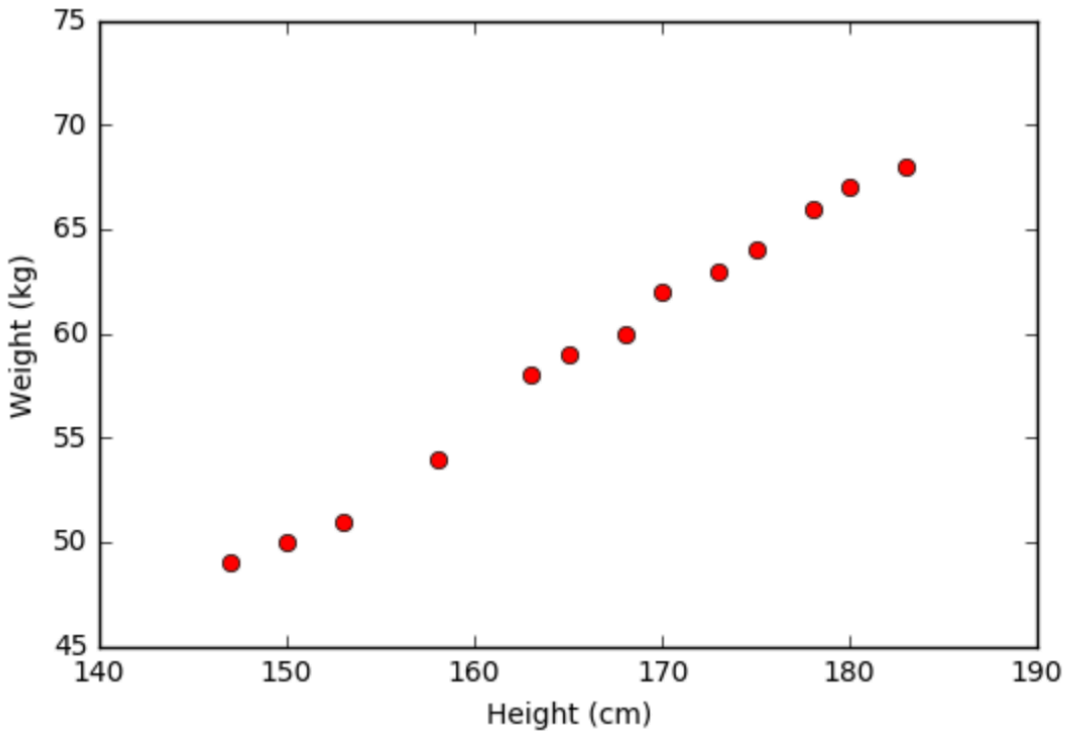
$$y = a*x + b$$

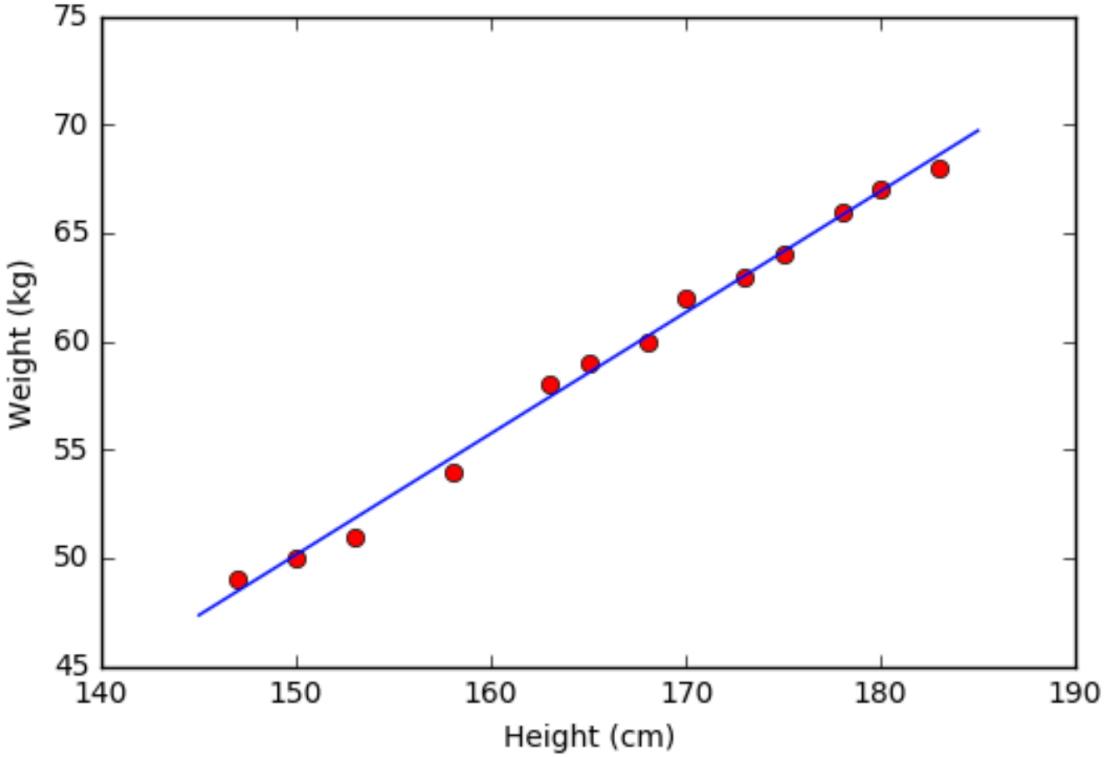
$$\text{Weight} = a*\text{Height} + b$$



Weight = w_1*Height + w_0

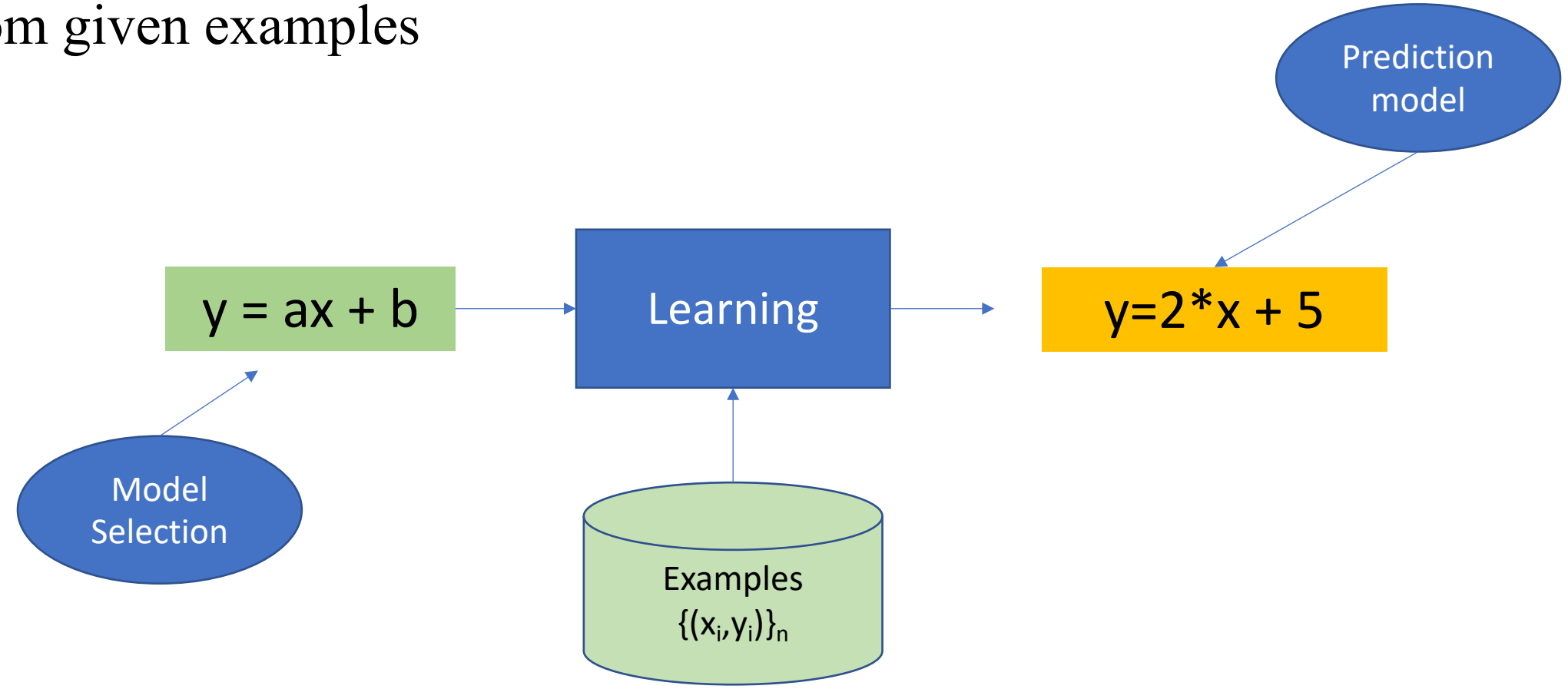
```
w =  [[-33.73541021]
 [  0.55920496]]
```





How to understand Learning?

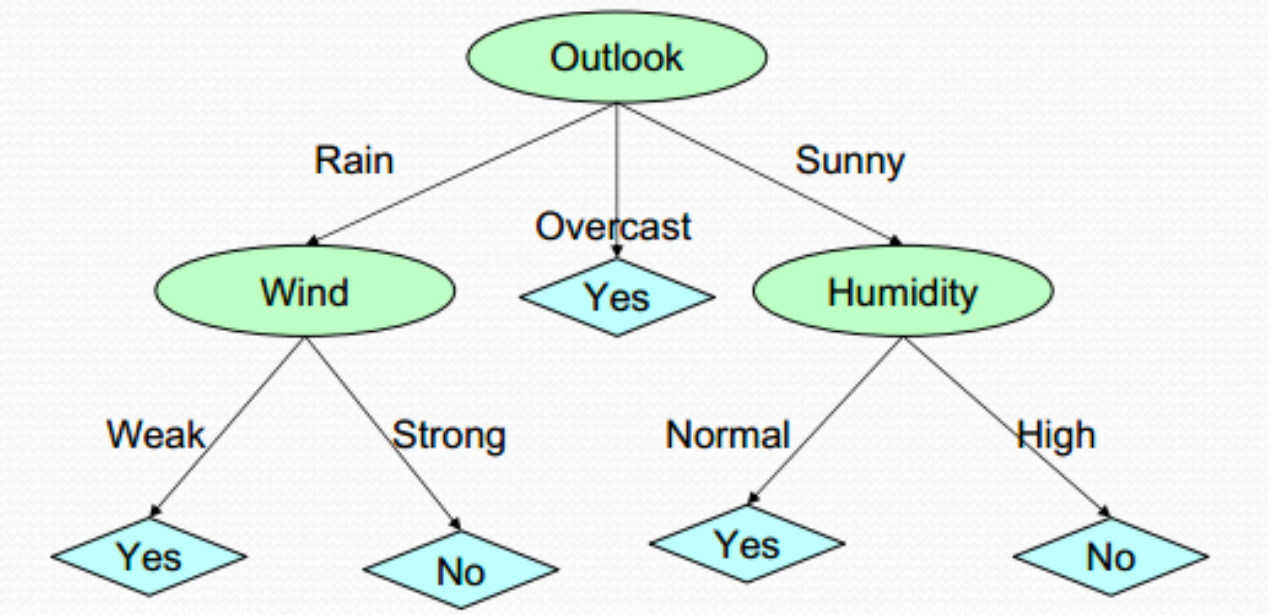
=> learn values for model's parameters by inferencing from given examples



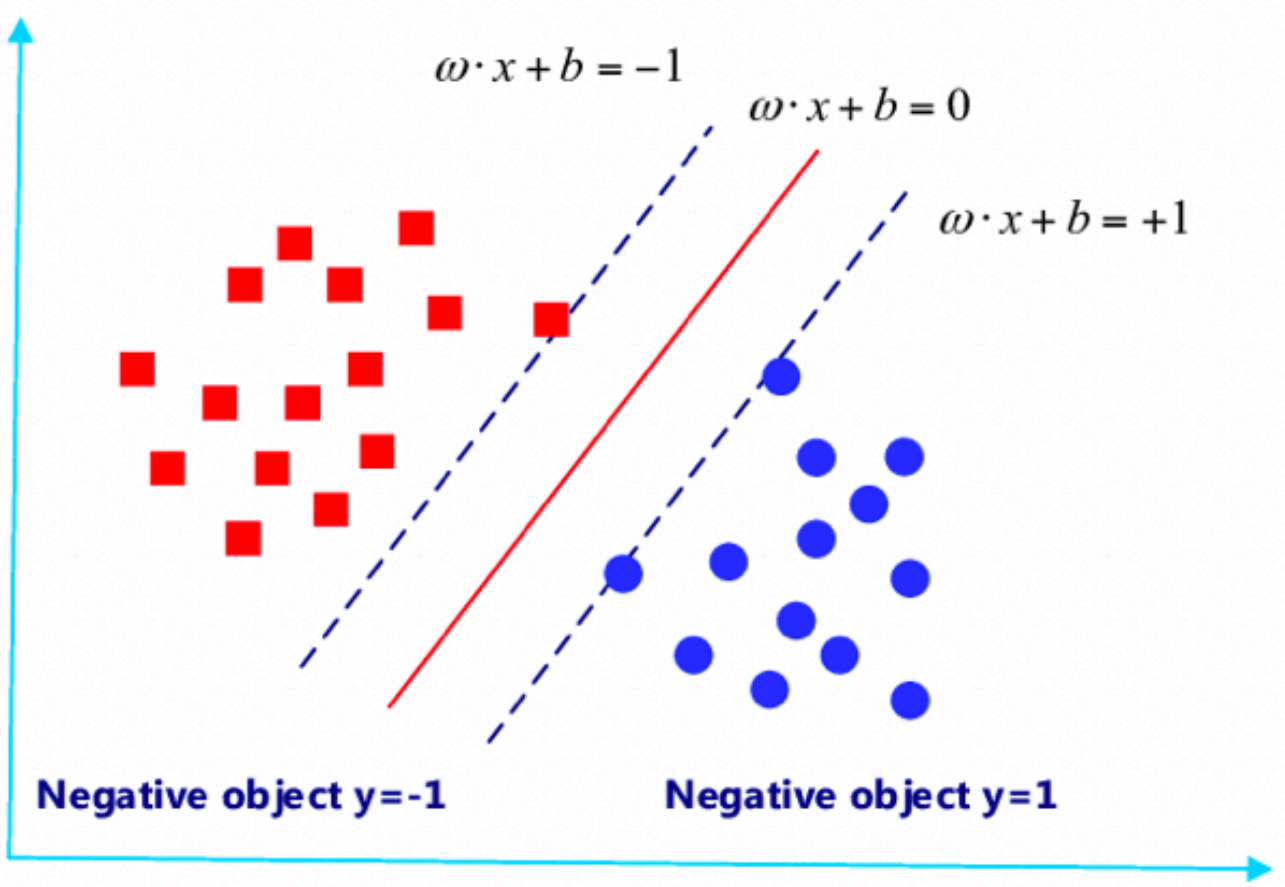
ML model: example 2

Outlook	Tempreature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

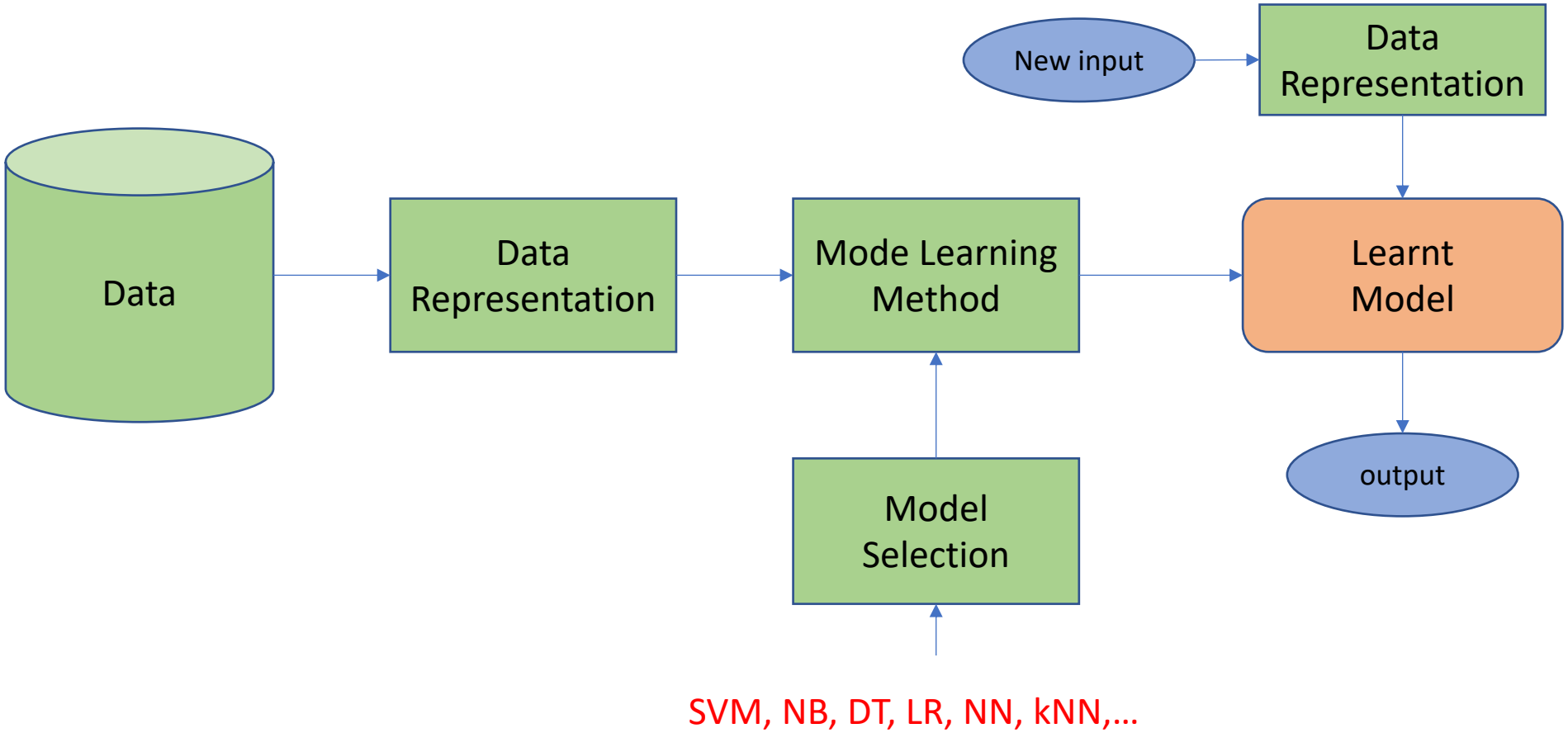
Decision Tree



ML model: example 3



A General Machine Learning Diagram



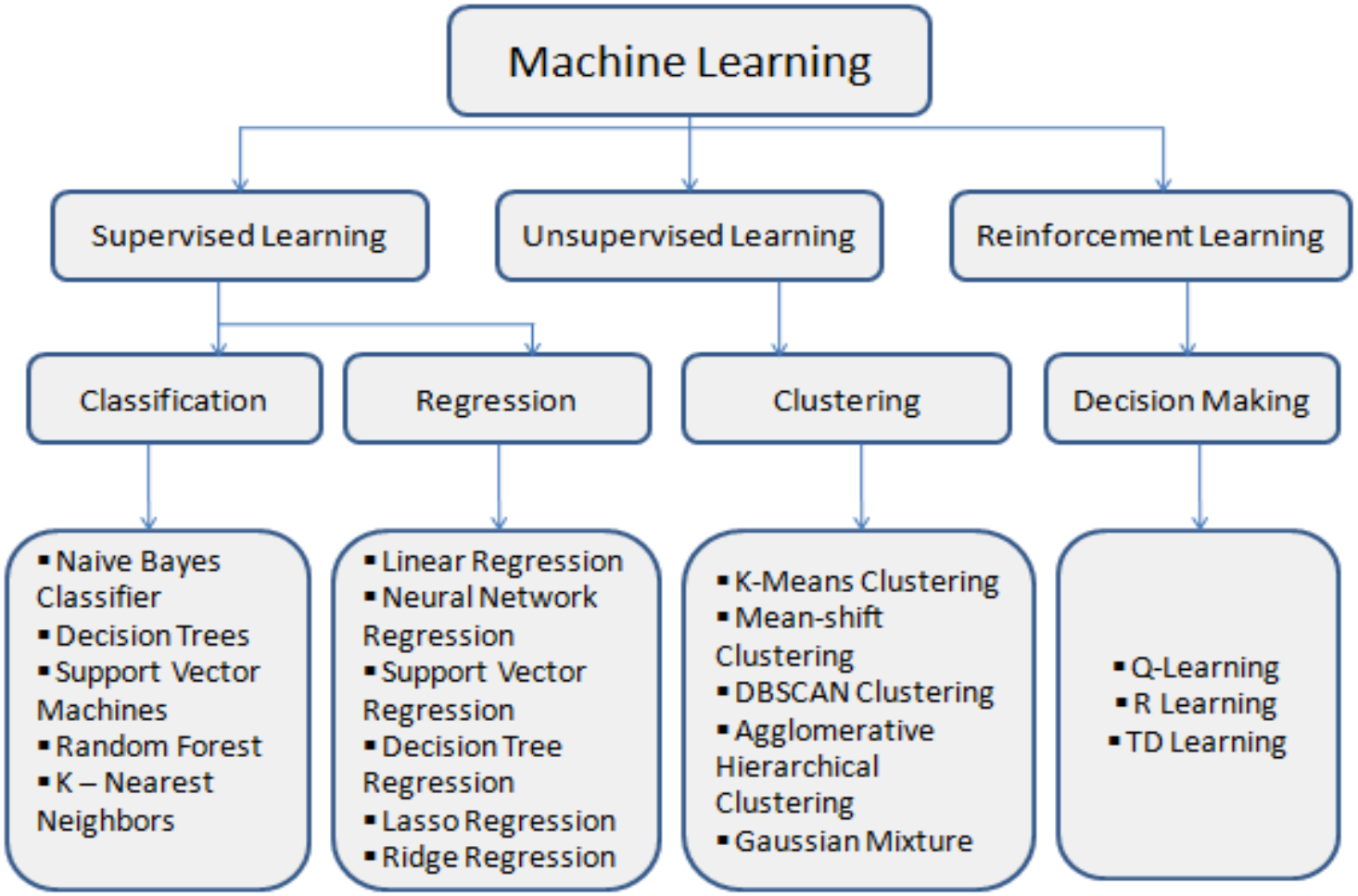
What should be learnt in Machine Learning?

- Understanding ML models
  - Architecture, Structure, or form of the models
  - Computation, inference/reasoning
- Methods/Algorithms for learning models' parameters
- Model evaluation
- Data Processing: data collection; feature selection; dimensional reduction, noisy filtering,…
- Other issues: overfitting,

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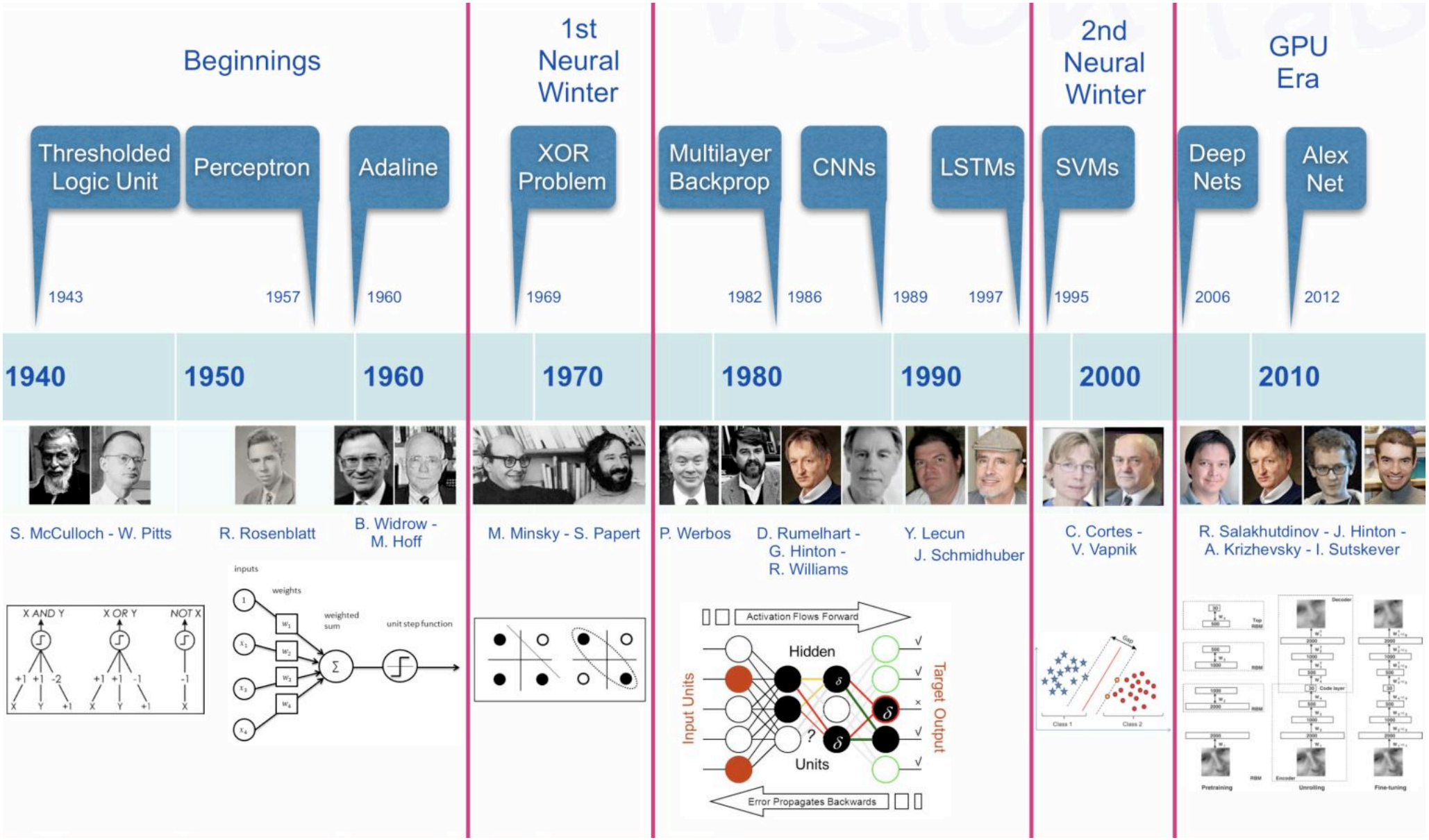
Types of Machine Learning



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History of ML



History of ML

Generative Model
(Very complicate)

ChatGPT,
Gemini,…
(Transformers,
Very Large
models)

Recognition
(complicated)

CNN, LSTM, GAN,…
(Bigdata, GPU)

Recognition
(Simple)

Perceptron, MLP,
NB, LR, SVM



1950 1980 2000 2010 2022

thousands of parameters millions of parameters billions of parameters

Topics of this course

1. Machine Learning: an Overview
2. Classification
    1. NB classifiers, Generative vs Discriminative models
    2. Decision Tree and Random Forest
    3. Support Vector Machine
    4. K-Nearest Neighbor
3. Regression
    1. Linear Regression and Gradient Descent Algorithm
    2. Logistic Regression for classification
4. Overfitting
5. Ensemble Learning
6. Unsupervised Learning
    1. Clustering
    2. Dimension Reduction
7. Neural Networks
    1. Feed Forward Neural Networks
    2. Recurrent Neural Network
8. Advanced Topics
    1. Deep Learning
    2. Reinforcement Learning

Course assessment

- Progress exercises (30%)
- Middle exam (20%)
  - Project and presentation
- Final exam (50%)
  - Project and presentation